

Active Learning for Learning-to-Defer

News topic classification on AG News with a simulated expert

Submitted by Group 29

1 Introduction

This project develops a news topic classifier that collaborates with an expert. For each article, the system either predicts the topic itself or defers the decision. The report covers a baseline classifier trained on all labels (Task 1), a simulated expert with uneven competence (Task 2), a learning-to-defer system that combines both predictors (Task 3), an active-learning study that collects a limited number of expert labels (Task 4), and an optional interface in which a real user supplies labels for actively selected articles (Task 5).

All experiments use the AG News dataset distributed through Hugging Face [1], with 120,000 training articles and 7,600 test articles, evenly split across World, Sports, Business, and Sci/Tech. The test split is used only for final evaluation. Reported values in the tables were transcribed from the JSON artifacts produced by the evaluation scripts, so the experiments can be reproduced by running the commands in Section 7.

2 Task 1: Baseline classifier

2.1 Experiment and design choices

The baseline is a TF-IDF representation followed by Logistic Regression, implemented with scikit-learn [2] and trained on all 120,000 labelled training articles. The vectoriser uses lowercasing, English stop-word removal, unigrams and bigrams, sublinear term-frequency scaling, a 100,000-feature limit, minimum document frequency 2, and maximum document frequency 0.95. The Logistic Regression classifier uses the lbfgs solver and a fixed random seed of 42.

This combination was chosen for three reasons. First, linear models with TF-IDF features are strong, well-understood baselines for short news texts. Second, Logistic Regression provides class-probability estimates that can be used as confidence scores and as compact features for the expert-competence model; these probabilities are not assumed to be perfectly calibrated. Third, this linear baseline can be reused to produce the probability features required by the later deferral experiments.

2.2 Results

The baseline reaches a test accuracy of 92.24%. Table 1 shows the per-class results. Sports has the highest F1-score (0.969), while Business (0.894) and Sci/Tech (0.902) have the lowest. The saved confusion matrix records 144 Business articles predicted as Sci/Tech and 109 Sci/Tech articles predicted as Business.

Class	Precision	Recall	F1-score	Support
World	0.937	0.911	0.924	1,900
Sports	0.956	0.982	0.969	1,900
Business	0.898	0.890	0.894	1,900
Sci/Tech	0.898	0.906	0.902	1,900

Table 1: Baseline test performance per class (accuracy 92.24%).

3 Task 2: Simulated expert

3.1 Design and justification

The simulated expert is a Business-news specialist. Its probability of answering correctly depends on the true topic: 97% for Business, 86% for World, 84% for Sci/Tech, and 70% for Sports. When wrong, it selects one of the other three classes. This creates complementary strengths in distinct regions of the label space. During deployment the deferral system does not observe the true label; it must infer competence indirectly from the baseline classifier's four probability outputs.

The expert is deterministic: correctness and the selected wrong label are derived from a hash of the article text and a fixed seed. Repeatedly querying the same article therefore produces the same answer, which makes the simulated-expert experiments exactly reproducible.

3.2 Results

The expert reaches an overall test accuracy of **84.26%**, clearly below the 92.24% baseline. Table 2 compares the designed and measured per-class accuracy; the small differences are ordinary sampling noise. The expert's strength (Business, 97.5%) is precisely where the baseline is weakest (89.0%), and its weakness (Sports, 71.0%) is where the baseline is strongest (98.2%). The two team members are therefore complementary, and a good deferral policy should be able to combine them into a team that beats both.

Class	Designed accuracy	Measured accuracy	Test samples
World	86.0%	85.9%	1,900
Sports	70.0%	71.0%	1,900
Business	97.0%	97.5%	1,900
Sci/Tech	84.0%	82.6%	1,900

Table 2: Simulated expert accuracy per class (overall 84.26%).

4 Task 3: Learning to defer

4.1 Method and design choices

In this task, simulated-expert answers are available for the complete training set. A Logistic Regression competence model predicts the probability that the expert is correct. Its inputs are the four class probabilities from the baseline classifier, and its binary target is 1 when the expert's training prediction matches the true training label. At test time, the system defers whenever estimated expert competence exceeds the AI's maximum class probability. The deferral margin is 0, so neither team member receives an additional preference.

A staged design was chosen instead of a joint learning-to-defer objective such as the consistent surrogate proposed by Mozannar and Sontag [3]. Joint optimisation is theoretically attractive, but the staged approach is simpler to interpret, reuses the Task 1 classifier unchanged, and can be retrained efficiently on the small query sets required in Task 4. This simplicity-versus-optimality trade-off is deliberate.

4.2 Results

The classifier-simulated-expert team reaches **94.84%** test accuracy, improving by **2.61** percentage points over the AI-only result of **92.24%** when calculated from the unrounded metric values, and by more than 10 points over the expert-only result of **84.26%**. The system defers **37.54%** of the test articles (2,853 of 7,600). An oracle that selects a correct team member whenever either one is correct reaches **99.17%**; this is a diagnostic upper bound, not a deployable policy, because it requires the unknown true label.

Metric	Value
AI-only accuracy	92.24%
Expert-only accuracy	84.26%
Team accuracy	94.84%
Oracle upper bound	99.17%
Deferral rate	37.5%
Accuracy on deferred articles (expert)	90.1%
Accuracy on non-deferred articles (AI)	97.7%
Beneficial deferrals (expert right, AI wrong)	424
Harmful deferrals (AI right, expert wrong)	226

Table 3: Learning-to-defer results on the test set (margin = 0).

Deferral quality is evaluated directly rather than only through team accuracy. There are **424** beneficial deferrals (expert correct, AI wrong) and **226** harmful deferrals (AI correct, expert wrong), for a net gain of **198** articles. Accuracy on non-deferred cases is **97.70%**, showing that the AI retains cases it handles reliably. The policy defers **73.4%** of Business articles, where the expert is strongest, but only **11.2%** of Sports articles, where the expert is weakest. Sports is the only class in which the team is less accurate than the AI; tuning a positive margin could reduce harmful deferrals at the cost of lower expert coverage.

Class	AI	Expert	Team	Deferral rate
World	91.1%	85.9%	94.0%	27.7%
Sports	98.2%	71.0%	96.7%	11.2%
Business	89.0%	97.5%	95.6%	73.4%
Sci/Tech	90.6%	82.6%	93.1%	37.8%

Table 4: Per-class accuracy and deferral behaviour of the team.

5 Task 4: Active learning for expert competence discovery

5.1 Setting and query strategy

Task 4 removes the assumption that expert answers are freely available. The classifier may still use all training labels, but expert answers exist only for articles that are explicitly queried, and the goal is to learn a good deferral policy with as few queries as possible.

The chosen strategy is deferral-boundary uncertainty sampling, following the general active-learning principle of querying informative examples [4]. Each run begins with **40** articles stratified across the four AI-predicted classes. The budgets in this report are **total cumulative budgets**: budget **50** contains the initial **40** plus 10 newly selected articles, while budget **2,000** contains exactly **2,000** expert labels. At every checkpoint the competence model scores unqueried articles, and active querying selects those for which estimated expert competence is closest to AI confidence. These are the cases near the defer-

or-answer boundary. Selection is performed in batches at total budgets **50**, 100, **250**, **500**, 1,000, and **2,000**, with the model retrained after each batch.

As a control, the same protocol is run with **random querying**. Every configuration is repeated with five random seeds (42, 52, 62, 72, 82), and all results are reported as mean and standard deviation over these five runs.

5.2 Metrics and results

Three kinds of metrics are reported. Team accuracy measures the final goal. ROC-AUC and the Brier score of the competence model on the test set measure how well expert competence itself has been learned (discrimination and calibration). The beneficial-deferral rate measures what fraction of deferrals actually help.

Budget	Random: team acc.	Active: team acc.	Random: AUC	Active: AUC
50	94.23% $\pm$ 0.52	94.66% $\pm$ 0.29	0.626	0.639
100	94.49% $\pm$ 0.05	94.70% $\pm$ 0.36	0.656	0.642
250	94.58% $\pm$ 0.12	94.71% $\pm$ 0.33	0.676	0.659
500	94.70% $\pm$ 0.23	94.86% $\pm$ 0.25	0.681	0.675
1,000	94.87% $\pm$ 0.11	94.96% $\pm$ 0.17	0.681	0.678
2,000	94.90% $\pm$ 0.07	94.97% $\pm$ 0.06	0.682	0.680

Table 5: Team accuracy (mean $\pm$ standard deviation over five seeds) and competence ROC-AUC for random and active querying.

Budget	Random: Brier	Active: Brier	Random: deferral / useful	Active: deferral / useful
50	0.130	0.131	33.11% / 16.46%	28.06% / 18.63%
100	0.127	0.131	34.40% / 15.92%	30.38% / 17.24%
250	0.127	0.130	35.90% / 15.37%	33.58% / 15.92%
500	0.126	0.128	36.40% / 15.40%	35.14% / 15.37%
1,000	0.125	0.127	35.97% / 15.39%	35.27% / 15.35%
2,000	0.125	0.127	36.14% / 15.34%	36.11% / 15.05%

Table 6: Competence Brier score and deferral / beneficial-deferral rates (mean over five seeds).

Three observations stand out. First, at a total budget of 50, active querying reaches **94.66% $\pm$ 0.29%** mean team accuracy, comparable to the **94.70% $\pm$ 0.23%** obtained by random querying at budget 500. This suggests strong low-budget query efficiency, but the overlapping run-to-run variation should not be interpreted as a formal tenfold guarantee. Second, the strategies converge as the budget grows. At 2,000 queries, active and random reach **94.97%** and **94.90%**, respectively. The active mean slightly exceeding the **94.84%** full-supervision result does not imply better global competence learning; boundary-focused sampling and run variation can improve the particular decision boundary. Third, active querying has higher AUC at budget 50, whereas random querying has slightly higher AUC from

budget 100 onward and slightly better Brier scores. Active querying instead gains its early team advantage through conservative, boundary-focused decisions: at budget 50 it defers **28.06%** rather than **33.11%** and has a higher beneficial-deferral rate (**18.63%** versus **16.46%**). This is consistent with sampling bias: boundary examples are useful for the team decision but are not representative of the full distribution.

6 Task 5 (optional): Interface with a human expert

The project is delivered as a Django web application with one page per task. In the optional Task 5 interface, a real user labels 20 articles taken from the last indices of the saved seed-42, 2,000-query active-learning sequence. These articles belong to the final active-query batch, but the current slicing procedure does not guarantee that they are the 20 smallest boundary-distance cases. Articles appear one at a time with four category choices. After submission, the page reveals the true topic, the AI prediction and confidence, estimated expert competence, the defer-or-use-AI decision, and the user's running accuracy. The answers currently support feedback and scoring only; feeding them back into the competence model would be required to close the real human-in-the-loop training cycle.

7 Reproducibility

All results can be regenerated with the following commands, run from the repository root in this order:

```
py -3.11 -m venv .venv
.\venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python -m learning_to_defer.services.train_baseline
python -m learning_to_defer.services.evaluate_expert
python -m learning_to_defer.services.evaluate_deferral
python -m learning_to_defer.services.evaluate_active_learning
python -m learning_to_defer.services.prepare_human_expert_pool
python manage.py migrate
python manage.py runserver
```

Each evaluation script writes metrics to JSON files in the artifacts directory, and the Django result pages read from those files. The Word/PDF report is a static document whose values were transcribed from those artifacts. The experiments use fixed random seeds and a deterministic simulated expert. The project should be run with Python 3.11 and the dependencies recorded in requirements.txt.

8 Conclusion

A simple, interpretable pipeline builds an effective classifier-simulated-expert team on AG News. The team reaches **94.84%**, exceeding the AI alone (**92.24%**) and the expert alone (**84.26%**) by deferring mainly where their strengths are complementary. With expert labels limited to **2,000** total queries (**1.67%** of the training set), active boundary sampling reaches **94.97%** mean team accuracy and is most valuable at small budgets. Important limitations remain: the competence model uses only four AI probabilities, the fixed zero margin permits harmful Sports deferrals, confidence calibration is not explicitly optimised, the simulated expert depends on the known class during evaluation, and Task 5 answers are not yet fed back into training. Richer text features, calibrated confidence, margin tuning, and a closed human-feedback loop are the strongest next steps.

9 References

- [1] X. Zhang, J. Zhao, and Y. LeCun, “Character-level Convolutional Networks for Text Classification,” Advances in Neural Information Processing Systems 28, 2015. Available: https://proceedings.neurips.cc/paper_files/paper/2015/hash/250cf8b51c773f3f8dc8b4be867a9a02-Abstract.html
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